

Homework 5

MATH 346/377 - Winter 2025

Due on Friday, March 28, 9:00pm on Crowdmark

Math 346: Do problems 1 to 5 – Total of points= 55 pts

Math 377: Do problems 1 to 6 – Total of points= 70 pts

1. (10 points) Find all integers $0 \leq x \leq 5^4 - 1 = 624$ such that

$$x^3 \equiv 3 \pmod{5^4}.$$

You can use Pari GP to help you with the computations, but you should show all the steps of your work.

Solution: First, we look at the solution modulo 5. We find that

$$1^3 \equiv 1 \pmod{5}, 2^3 \equiv 3 \pmod{5}, 3^3 \equiv 2 \pmod{5}, 4^3 \equiv 4 \pmod{5}.$$

Hence modulo 5 we have a unique solution $x_0 = 2$. The derivative of $f(x) = x^3 - 3$ is $f'(x) = 3x^2$ and $f'(2) \equiv 2 \not\equiv 0 \pmod{5}$ so that we can lift the solution x_0 to a unique solution x_e modulo 5^e for all $e \geq 2$.

To lift the solution we want to find the unique $0 \leq a \leq 4$ such that $0 \leq x_1 = 2 + 5a \leq 5^2 - 1$ satisfies $x_1^3 - 3 \equiv 0 \pmod{5^2}$. In that case, we should have

$$0 \equiv (2 + 5a)^3 - 3 \equiv (2^3 + 3 \cdot 2^2 \cdot 5a) - 3 \equiv 5 + 3 \cdot 2^2 \cdot 5a \pmod{5^2}.$$

Since the RHS is divisible by 5, this also implies that

$$0 \equiv 1 + 3 \cdot 2^2 a \equiv 1 + 2a \pmod{5}.$$

We deduce that $a = 2$ and $x_1 = 2 + 2 \cdot 5 = 12$ is the lift of x_0 modulo 5^2 .

To lift x_1 to a solution x_2 modulo 5^3 we repeat the process. Set $x_2 = 12 + 5^2 b$ and want to find unique $0 \leq b \leq 4$ such that $x_2^3 \equiv 3 \pmod{5^3}$. As above we find that

$$b \equiv \frac{3 - 12^3}{5^2} \frac{1}{3 \cdot 12^2} \equiv 3 \pmod{5},$$

where $\frac{3-12^3}{5^2} = -69 \equiv 1 \pmod{5}$. Thus, we find that $x_2 = 12 + 5^2 \cdot 3 = 87$. Repeating this one more time with $x_3 = 87 + 5c$, we find that

$$c \equiv \frac{3 - 87^3}{5^3} \cdot \frac{1}{387^2} \equiv 1 \pmod{5},$$

where $\frac{3-87^3}{5^3} = -5268 \equiv 3 \pmod{5}$. Hence, the only solution to $x^3 \equiv 3 \pmod{5^4}$ such that $0 \leq x_3 \leq 624$ is

$$x_3 = 87 + 5^3 = 212.$$

2. (13 points) (a) (3 points) Show that the p -adic expansion of $\frac{1}{1-p}$ is

$$\frac{1}{1-p} = \sum_{i=0}^{\infty} p^i = 1 + p + p^2 + p^3 + p^4 + \cdots$$

- (b) (4 points) Find the p -adic expansion of $x = \frac{1+p}{1-p}$.
(c) (6 points) Suppose that $x \in \mathbb{Z}_p$ has p -adic expansion $x = \sum_{i=0}^{\infty} x_i p^i$. What is the p -adic expansion of $-x$? In particular, what is the p -adic expansion of -1 ?

Solution:

- (a) We have seen that we can find the p -adic expansion of rational number by computing the successive residues modulo p^n . For $1/(1-p)$ we can use $(1-p^n) = (1-p)(1+p+\cdots+p^{n-1})$ to show that

$$\frac{1}{1-p} \equiv 1 + p + \cdots + p^{n-1} \pmod{p^n},$$

so as n goes to infinity this gives the desired p -adic expansion.

- (b) Using the p -adic expansions of $p+1$ and $\frac{1}{1-p}$ we find that

$$\begin{aligned} (1+p) \frac{1}{1-p} &= (1+p)(1+p+p^2+p^3+\cdots) \\ &= (1+p) + p(1+p) + p^2(1+p) + \cdots \\ &= 1 + 2p + 2p^2 + 2p^3 + \cdots \\ &= 1 + \sum_{i=1}^{\infty} 2p^i. \end{aligned}$$

Note that when $p = 2$ the 2-adic expansion is

$$1 + \sum_{i=1}^{\infty} 2^{i+1} = 1 + 2^2 + 2^3 + 2^4 + \dots \quad (1)$$

Alternatively, we can use the method seen in class to find the p -adic expansion of a rational number by computing its residues modulo p^n :

$$\begin{aligned} \frac{1+p}{1-p} &\equiv \frac{(1+p)(1+p+\dots+p^{n-1})}{1-p^n} \\ &\equiv (1+p)(1+p+\dots+p^{n-1}) \\ &\equiv 1+2p+2p^2+\dots+2p^{n-1} \pmod{p^n} \end{aligned}$$

where we used that $p^n - 1 = (p-1)(1+p+\dots+p^{n-1})$. As $n \rightarrow \infty$, this gives the same p -adic expansion as above.

- (c) Without loss of generality we can assume that $x_0 \neq 0$ (If $x_0 = 0$, then $x = p^i x'$ for some $x' = x'_0 + p\mathbb{Z}_p$ with $x'_0 \neq 0$ and the result will follow from the p -adic expansion of x'). We have that

$$-x = -x_0 - x_1p - x_2p^2 - x_3p^3 - \dots$$

with $1 \leq x_0 \leq p-1$ and $0 \leq x_i \leq p-1$ for $i \geq 1$. This is not the p -adic expansion because the coefficients $-x_i$ are negative. Since $-x_0 \equiv p-x_0 \pmod{p}$, we replace $-x_0$ by $p-x_0$ and add a $-p$ to compensate. We get

$$-x = (p-x_0) - (1+x_1)p - x_2p^2 - x_3p^3 - \dots$$

The first coefficient satisfies $1 \leq p-x_0 \leq p-1$, since $1 \leq x_0 \leq p-1$. However, the second coefficient $-(1+x_1)$ is a priori still negative, so we repeat the process and add $+p^2 - p^2$ to get

$$-x = (p-x_0) + (p-1-x_1)p - (x_2+1)p^2 - x_3p^3 - \dots$$

The second coefficient satisfies $0 \leq p-1-x_1 \leq p-1$, but since the third is still negative we add $+p^3 - p^3$. Repeating this process gives the p -adic expansion of $-x$

$$-x = (p-x_0) + \sum_{i=1}^{\infty} (p-1-x_i)p^i.$$

In particular, since the p -adic expansion of 1 is $1 + 0 \cdot p + 0 \cdot p^2 + \dots$, the p -adic expansion of -1 is

$$-1 = (p-1) + (p-1)p + (p-1)p^2 + \dots$$

This can also be seen from the following

$$-1 = \frac{p-1}{1-p} = (p-1)(1+p+p^2+p^3+\dots).$$

3. (10 points) Let $x = -\frac{2}{3}$ and $p = 5$.

- (a) (5 points) What is the 5-adic norm of x ? Compute the first few terms of the 5-adic expansions of x . Based on your observations, make a conjecture about the 5-adic expansion of x .

You can verify your solution using the Pari GP command `print(x+O(p^k))`, which gives the first $k - 1$ terms in the p -adic expansion of x .

- (b) (5 points) Prove your conjecture.

Solution:

- (a) We have $|- \frac{2}{3}|_5 = 1$, so $- \frac{2}{3} \in \mathbb{Z}_5$. We have

$$x = -\frac{2}{3} \equiv -4 \equiv 1 \pmod{5}$$

so that $x = 1 + 5q_1$, where $0 \leq q_1 \leq 4$ satisfies

$$q_1 = \frac{x - 1}{5} = -\frac{1}{3} \equiv 3 \pmod{5}.$$

Hence $x = 1 + 3 \cdot 5 + q_2 5^2 = 16 + q_2 5^2$, where $0 \leq q_2 \leq 4$ satisfies

$$q_2 = \frac{x - 16}{5^2} = -\frac{50}{3 \cdot 5^2} = -\frac{2}{3} \equiv 1 \pmod{5}.$$

So $x = 1 + 3 \cdot 5 + 5^2 + \dots$. Repeating this process shows that

$$x = 1 + 3 \cdot 5 + 5^2 + 3 \cdot 5^3 + 5^4 + 3 \cdot 5^5 + \dots$$

We can conjecture that the 5-adic expansion is 1313131313... has period 2 with repeating sequence 13.

- (b) Let x be the 5-adic integer with the 5-adic expansion 1313131313..., *i.e.* with period 2 and repeating sequence 13. By grouping all the exponents with coefficients 1 respectively 3, we get

$$\begin{aligned} x &= (1 + 5^2 + 5^4 + \dots) + 3(5 + 5^3 + 5^5 + \dots) \\ &= \frac{1}{1 - 5^2} + 3 \cdot 5 \frac{1}{1 - 5^2} \\ &= \frac{1 + 3 \cdot 5}{1 - 5^2} \\ &= -\frac{16}{24} \\ &= -\frac{2}{3}. \end{aligned}$$

This confirms our conjecture from part (a).

A n -th root of unity is a root of the polynomial $x^n - 1$. In $\mathbb{C}$, all the n -th roots of unity are given by $e^{\frac{2i\pi k}{n}}$ with $0 \leq k \leq n-1$. When $p = 5$, we have seen in class that the 4-th roots of unity $\pm 1, \pm i$ are contained in $\mathbb{Z}_5$. The next problem generalizes this to other primes.

4. (12 points) (a) (4 points) Use Hensel's Lemma to show that $\mathbb{Z}_p$ contains $p-1$ distinct $(p-1)$ -th roots of unity μ_r , each congruent to a unique $1 \leq r \leq p-1$.
 (b) (4 points) The roots can also be described in a different way. For an integer r coprime to p , show that the sequence $x_n = r^{p^n}$ converges with respect to the p -adic norm, and that

$$\mu_r := \lim_{n \rightarrow \infty} r^{p^n}$$

is the $(p-1)$ -th root of unity that is congruent to r modulo p .

- (c) (4 points) Let $p = 7$ and $r = 2$. Use Newton's method to compute the first few terms of the 7-adic expansion of μ_2 , and compare it with the 7-adic expansion of $\lim_{n \rightarrow \infty} 2^{7^n}$ given by Pari GP.

For Newton's method you should use Pari GP to help you with the computations. You can define polynomials simply by writing `f(x)=x^6-1` for example. On the other hand you will have a good approximation of the limit of 2^{7^n} by taking 2^{7^8} . The first five nine terms are given by the command `print(2^(7^8)+O(7^10))`.

Solution:

- (a) Let $f(x) = x^{p-1} - 1$ and $f'(x) = (p-1)x^{p-2}$. By Fermat's little theorem, any $1 \leq r \leq p-1$ is a root of $f(x) \equiv 0 \pmod{p}$, and $f'(r) \not\equiv 0 \pmod{p}$. By Hensel's Lemma, each r lifts to a unique $\mu_r \in \mathbb{Z}_p$ such that $\mu_r = r + p\nu_r$ for some $\nu_r \in \mathbb{Z}_p$. This gives us the $p-1$ distinct roots of unity.

- (b) We use the criterion seen in class, that the sequence x_n converges if and only if $\lim_{n \rightarrow \infty} |x_{n+1} - x_n|_p = 0$. If r is coprime to p , then by Fermat's little theorem we have

$$r^{p^{n+1}} = r^{p^n(p-1)} r^{p^n} = r^{\phi(p^{n+1})} r^{p^n} \equiv r^{p^n} \pmod{p^{n+1}}.$$

Hence, we have

$$0 \leq |x_{n+1} - x_n|_p = |r^{p^{n+1}} - r^{p^n}|_p \leq \frac{1}{p^{n+1}},$$

so that $\lim_{n \rightarrow \infty} |x_{n+1} - x_n|_p = 0$.

Moreover, we have that

$$x_n^{p-1} = r^{p^n(p-1)} \equiv 1 \pmod{p^{n+1}},$$

so that $|x_n^{p-1} - 1|_p \leq \frac{1}{p^{n+1}}$. Thus, we have

$$|\mu_r^{p-1} - 1|_p = \lim_{n \rightarrow \infty} |x_n^{p-1} - 1|_p = 0$$

and $\mu_r^{p-1} = 1$ (since $|x|_p = 0$ implies that $x = 0$). This shows that μ_r is a $(p-1)$ -th root of unity.

- (c) Newton's method gives us a sequence of solutions $0 \leq x_n \leq 7^{n+1} - 1$ to $f(x_n) = x_n^6 - 1 \equiv 0 \pmod{7^{n+1}}$ each satisfying $x_n \equiv x_0 \pmod{7}$. The sequence starts at $x_0 = 2$ and the higher terms are given by

$$x_n = x_{n-1} + q_n 7^n, \quad q_n := -\frac{f(x_{n-1})}{f'(x_{n-1}) 7^n} = -\frac{x_{n-1}^6 - 1}{6x_{n-1}^5 7^n} \pmod{7}.$$

For the first few terms we find that

$$\begin{aligned} x_1 &= 2 + q_1 7 = 30, & q_1 &= -\frac{2^6 - 1}{6 \cdot 2^5 7} \equiv 4 \pmod{7} \\ x_2 &= 30 + q_2 7^2 = 324, & q_2 &= -\frac{30^6 - 1}{6 \cdot 30^5 7^2} \equiv 6 \pmod{7} \\ x_3 &= 324 + q_3 7^3 = 1353, & q_3 &= -\frac{324^6 - 1}{6 \cdot 324^5 7^3} \equiv 3 \pmod{7} \\ x_4 &= 1353 + q_4 7^4 = 1353, & q_4 &= -\frac{1353^6 - 1}{6 \cdot 1353^5 7^4} \equiv 0 \pmod{7} \\ x_5 &= 1353 + q_5 7^5, & q_5 &= -\frac{1353^6 - 1}{6 \cdot 1353^5 7^5} \equiv 2 \pmod{7}. \end{aligned}$$

We find the first five terms of the p -adic expansion

$$\mu_2 = 2 + 4 \cdot 7 + 6 \cdot 7^2 + 3 \cdot 7^3 + 2 \cdot 7^5 + O(7^6)$$

On the other hand, the command `print(2^(7^8)+0(7^6))` returns the first five terms

$$2 + 4 \cdot 7 + 6 \cdot 7^2 + 3 \cdot 7^3 + 2 \cdot 7^5 + O(7^6)$$

5. (10 points) Show that the polynomial

$$f(x) = (x^2 - 13)(x^2 - 17)(x^2 - 221)$$

has a root in $\mathbb{R}$ and $\mathbb{Q}_p$ for all odd primes¹, but not in $\mathbb{Q}$.

Hint: Quadratic reciprocity.

Solution: Over $\mathbb{R}$ the polynomial has roots $\pm\sqrt{3}, \pm\sqrt{13}$ and $\pm\sqrt{39}$, and it is clear that none of these roots are in $\mathbb{Q}$.

For the roots over $\mathbb{Q}_p$, it is enough to show that for each odd prime p , one of the three polynomials $(x^2 - 3), (x^2 - 13)$ or $(x^2 - 39)$ has a root in $\mathbb{Q}_p$. Note that $13 \cdot 17 = 221$.

When $p = 13$ the polynomial $x^2 - 17 \equiv x^2 - 4 \pmod{13}$ has a root $x_0 = 2$ modulo 13, which can be lifted to a root in $\mathbb{Q}_{13}$ by Hensel's Lemma.

When $p = 17$, the polynomial $x^2 - 13$ has a root modulo 17. Indeed, 13 is a quadratic residue modulo 17 since

$$\begin{aligned} \left(\frac{13}{17}\right) &= \left(\frac{17}{13}\right) (-1)^{\frac{17-1}{2} \frac{13-1}{2}} \\ &= \left(\frac{17}{13}\right) \\ &= \left(\frac{4}{13}\right) \\ &= 1. \end{aligned}$$

By Hensel's Lemma again, this root modulo 17 can be lifted to a root in $\mathbb{Q}_{17}$.

Finally, for all other odd primes $p \neq 13, 17$ we have

$$\left(\frac{13}{p}\right) \left(\frac{17}{p}\right) = \left(\frac{13 \cdot 17}{p}\right) = \left(\frac{221}{p}\right).$$

This implies one of the three Legendre symbols

$$\left(\frac{13}{p}\right), \quad \left(\frac{17}{p}\right), \quad \left(\frac{221}{p}\right)$$

must be equal to 1. This implies that one of the three polynomials

$$(x^2 - 3), \quad (x^2 - 13), \quad (x^2 - 39)$$

has a root modulo p , which can then be lifted to a root in $\mathbb{Q}_p$.

¹For $p = 2$, the version of Hensel's Lemma that we have seen in class does not allow us to apply it to find roots of quadratic polynomials in $\mathbb{Q}_2$, since the derivative of $x^2 - a$ is $2x$, hence not coprime to 2. However Hensel's Lemma can be strengthened to make it work here, and show that $x^2 - 17$ has a root in $\mathbb{Q}_2$. Thus, one can deduce that $f(x)$ has root in $\mathbb{R}$ and $\mathbb{Q}_p$ for *any* p , so it is an example of a polynomial that does not satisfy the local-global principle.

Problem 6 for MATH 377 students.

6. (15 points) Let $x \in \mathbb{Q}_p$ such that $|x|_p = 1$. The goal of this problem is to show that the p -adic expansion of x is periodic if and only if $x \in \mathbb{Q}$ and $-1 \leq x < 0$.

Remark: More generally, it is true that the p -adic expansion of $x \in \mathbb{Q}_p$ is eventually periodic if and only if $x \in \mathbb{Q}$.

- (a) (5 points) Show the first implication: if the p -adic expansion of x is periodic, then x is rational and $-1 \leq x < 0$.
- (b) (6 points) Show that if there is an integer $n \geq 1$ such that $(1 - p^n)x$ is an integer between and

$$0 < (1 - p^n)x \leq p^n - 1,$$

then the p -adic expansion of x is periodic.

- (c) (4 points) Use part b) to deduce the other direction: if $x \in \mathbb{Q}$ is such that $|x|_p = 1$ and $-1 \leq x < 0$, then x is periodic.

Solution:

- (a) Let $x \in \mathbb{Q}_p$ be such that $|x|_p = 1$ and with periodic p -adic expansion

$$x = x_0 + px_1 + p^2x_2 + \cdots \quad x_0 \neq 0.$$

(Note that we have $x_0 \neq 0$ because $|x|_p = 1$). Since x is periodic we have

$$x = (x_0 + x_1p + \cdots + x_{n-1}p^{n-1}) + (x_0p^n + x_1p^{n+1} + \cdots) + \cdots$$

for some n . By grouping all the terms with the same coefficient, we can write

$$\begin{aligned} x &= (x_0 + px_1 + \cdots + p^{n-1}x_{n-1}) (1 + p^n + p^{2n} + p^{3n} + \cdots) \\ &= \frac{x_0 + px_1 + \cdots + p^{n-1}x_{n-1}}{1 - p^n} \\ &= -\frac{x_0 + px_1 + \cdots + p^{n-1}x_{n-1}}{p^n - 1}. \end{aligned}$$

Hence $y \in \mathbb{Q}$. Moreover, since $0 < x_0 \leq p - 1$ and $0 \leq x_i \leq p - 1$ for $i > 0$, we have $x < 0$ and

$$-x = \frac{x_0 + px_1 + \cdots + p^{n-1}x_{n-1}}{p^n - 1} \leq (p - 1) \frac{1 + p + \cdots + p^{n-1}}{p^n - 1} = 1.$$

Thus we have $-1 \leq x < 0$

(b) Since $|x|_p = 1$, x is a p -adic integer with p -adic expansion

$$x = x_0 + px_1 + p^2x_2 + \cdots$$

where $x_0 \neq 0$. Then

$$p^n x = x_0 p^n + p^{n+1}x_1 + p^{n+2}x_2 + \cdots .$$

and

$$(1 - p^n)x = x_0 + x_1p + \cdots + x_{n-1}p^{n-1} + \sum_{i=n}^{\infty} (x_i - x_{i-n})p^i.$$

Hence if $(1 - p^n)x$ is an integer less than $p^n - 1$, we see that its p -adic expansion cannot contain any powers of p larger than p^{n-1} and must be

$$(1 - p^n)x = x_0 + x_1p + \cdots + x_{n-1}p^{n-1}.$$

It follows from the above expansion that $x_i - x_{i-n} = 0$ for all $i \geq n$, which shows that $x_j = x_{j+n}$ for all $j \geq 0$. This shows that the sequence is periodic.

(c) Let $x = \frac{a}{b}$ with $-1 \leq x < 0$, where the fraction is reduced. We know that b is coprime to p , otherwise we would not have $x \in \mathbb{Z}_p$. Hence, there is an n such that $p^n - 1 \equiv 0 \pmod{b}$ and $(1 - p^n)x = (1 - p^n)\frac{a}{b}$ is an integer. Moreover, since $-1 < x \leq 0$, we have

$$0 \leq (1 - p^n)x \leq p^n - 1.$$

It follows from the previous part that x is periodic.